Section 5.2 Conditional probability & independence

条件概率与独立性

Method 1 / 方法 1 [PS02-M01]

Information changes the probability space. Keep prior weights, evidence likelihoods, conditional cases and independence assumptions explicit; the formulas are short once the correct restricted or state-based model is chosen.

信息会改变概率空间。明确写出先验权重、证据似然、条件分类与独立假设；选对受限模型或状态模型后，公式本身通常很短。

Renormalize the sample space after information is given

给定信息后重新归一样本空间

Procedure

1. Name the conditioning event B and require $P(B) > 0$.

2. Use $P(A \mid B) = P(A \cap B)/P(B)$; in a uniform model count only outcomes surviving B.

3. Express structural information through gaps, degrees, repeated positions or disjoint Venn regions before counting.

4. Check the restricted outcomes remain equally likely; do not inherit independence without proof.

通用步骤

1. 命名条件事件 B，并要求 $P(B) > 0$。

2. 使用 $P(A \mid B) = P(A \cap B)/P(B)$；均匀模型中只统计条件 B 下保留的结果。

3. 计数前先用间隙、度数、重复位置或互斥 Venn 区域表达结构信息。

4. 检查受限结果仍等概率，不能未经证明沿用独立性。

Section 5.2 Conditional probability & independence

条件概率与独立性

Method 1 / 方法 1 [PS02-M01] (continued / 续)

> Example. Two fair dice are rolled. Given their sum is even, find the chance that both are odd.

> 例题：投两枚公平骰子，已知和为偶数，求两枚都为奇数的概率。

Worked solution

1. Even sum means both odd or both even. Each class has $3 \cdot 3 = 9$ ordered outcomes.

2. The restricted sample space therefore has eighteen equally likely outcomes, with nine favorable.

3. The conditional probability is $1/2$. The oddness events are dependent after conditioning, since knowing one determines the other.

例题解析

1. 和为偶数表示同奇或同偶，每类均有 $3 \cdot 3 = 9$ 个有序结果。

2. 条件样本空间有十八个等可能结果，其中九个有利。

3. 条件概率为 $1/2$；条件下两个奇偶事件相关，因为知道一个就确定另一个。

Conditional denominators count the information event, not the original whole space.

条件概率的分母统计已知信息事件，而非原整个空间。

Section 5.2 Conditional probability & independence

条件概率与独立性

Method 1 / 方法 1 [PS02-M01] (continued / 续)

For a uniformly chosen bipartite edge set covering all vertices with one extra edge, the repeated degree-two row and column control whether an edge is removable. Prove uniqueness of the removable edge before using a matching-plus-extra-edge count; otherwise that construction can overcount.

Visibility of a sticker attached at time j means that no later roll uses its face. Convert each visibility statement into forbidden equalities of roll positions, then count the allowed sequences under all conditions simultaneously.

For conditioned runs in a random lineup, place blocks in the gaps around the other group and enumerate every ordered composition of the block sizes. Cancel common internal-label factors only after verifying they are shared by all retained patterns.

均匀选取的二部边集若覆盖全部顶点且只多一条边，重复的二度行与列决定是否有边可删。用“匹配加一条边”计数前，须证明可删边唯一，否则该构造会重复计数。

时刻 j 贴上的贴纸最终可见，表示后续投掷都未使用其所在面。把每个可见声明转为投掷位置之间被禁止的相等关系，再同时满足全部条件计数。

随机队列的条件连续段问题，可将各块放入另一组周围的间隙，并枚举全部有序块长拆分。只有确认保留模式共享相同类内标签因子后，才能将它们约去。

Practice / 配套练习: AIME-2010-II-Q13 (WB p. 764), AIME-2011-I-Q12 (WB p. 765), AIME-2014-II-Q02 (WB p. 752), AIME-2015-II-Q02 (WB p. 754), AIME-2017-II-Q09 (WB p. 762), AIME-2024-I-Q04 (WB p. 760), AIME-2026-I-Q09 (WB p. 763)

Section 5.2 Conditional probability & independence

条件概率与独立性

Method 2 / 方法 2 [PS02-M02]

Update a hidden model with Bayes before predicting again

再次预测前先用贝叶斯更新隐藏模型

Procedure

1. List the possible hidden models H_i and their prior probabilities.

2. Multiply each prior by the likelihood of the observed evidence and normalize the resulting weights.

3. Average the next-event probability using these posterior weights; conditional independence given a model need not survive averaging over models.

通用步骤

1. 列出可能隐藏模型 H_i 及先验概率。

2. 将每个先验乘观测证据的似然，再归一所得权重。

3. 以后验权重平均下一事件概率；给定模型的条件独立，未必在对模型平均后仍成立。

> Example. Choose equally between a fair coin and an always-heads coin, then observe one head. Are the next two tosses independent given that observation?

> 例题：等可能选择公平硬币或总为正面的硬币，观察到一次正面后，接下来两次投掷在此条件下独立吗?

Worked solution

1. The evidence likelihoods are $1/2$ and one, so posterior probabilities are fair $1/3$, always-heads $2/3$.

Section 5.2 Conditional probability & independence

条件概率与独立性

Method 2 / 方法 2 [PS02-M02] (continued / 续)

2. Each future toss is heads with probability $(1/3)(1/2) + (2/3) = 5/6$.

3. Two future heads have probability $(1/3)(1/4) + (2/3) = 3/4$.

4. Since $3/4 \neq (5/6)^2$, the future tosses are dependent under the observed-information condition.

例题解析

1. 证据似然分别为 $1/2$ 与一，故后验为公平币 $1/3$、总正面币 $2/3$。

2. 每次未来投掷为正面的概率为 $(1/3)(1/2) + (2/3) = 5/6$。

3. 未来两次均正面的概率为 $(1/3)(1/4) + (2/3) = 3/4$。

4. 因 $3/4 \neq (5/6)^2$，在已观察信息条件下，两次未来投掷相关。

Evidence changes the model weights even when the physical die or coin remains unchanged.

即使实际骰子或硬币不变，证据仍会改变各模型权重。

Practice / 配套练习: AIME-2014-II-Q06 (WB p. 761)

Section 5.2 Conditional probability & independence

条件概率与独立性

Method 3 / 方法 3 [PS02-M03]

Condition on a complete first-stage partition

按完整的第一阶段分类取条件

Procedure

1. Choose disjoint cases H_i whose union is the whole space, such as opponents, bracket pairings or geometric regions.

2. Compute $P(A \mid H_i)$ within each case, using the probabilities that hold there.

3. Combine by $P(A) = \sum_i P(H_i)P(A \mid H_i)$. This also gives an equation for unknown conditional rates.

4. If coordinates are claimed independent after conditioning, identify the product region or independent coordinate variables that justify it.

通用步骤

1. 选择互斥且并为全空间的情形 H_i，例如对手、赛程配对或几何区域。

2. 在各情形内计算 $P(A \mid H_i)$，使用该情形实际成立的概率。

3. 以 $P(A) = \sum_i P(H_i)P(A \mid H_i)$ 合并，也可由此求未知条件概率。

4. 若声称取条件后坐标独立，须指出相应乘积区域或独立坐标变量。

> Example. A mode is selected: mode A with probability $1/4$, otherwise mode B. Success rates are $1/2$ and $1/3$. Find the overall success probability and the posterior chance of mode A after success.

Section 5.2 Conditional probability & independence

条件概率与独立性

Method 3 / 方法 3 [PS02-M03] (continued / 续)

例题：先选模式：概率 $1/4$ 选 A，否则选 B；成功率分别为 $1/2, 1/3$。求总体成功率及成功后为模式 A 的概率。

Worked solution

1. The disjoint success contributions are $(1/4)(1/2) = 1/8$ and $(3/4)(1/3) = 1/4$.

2. The total success probability is $1/8 + 1/4 = 3/8$.

3. Of that success mass, mode A contributes $1/8$, so its posterior probability is $(1/8)/(3/8) = 1/3$.

例题解析

1. 两种互斥成功贡献为 $(1/4)(1/2) = 1/8$ 与 $(3/4)(1/3) = 1/4$。

2. 总成功率为 $1/8 + 1/4 = 3/8$。

3. 成功质量中模式 A 贡献 $1/8$，故其后验概率为 $(1/8)/(3/8) = 1/3$。

Use the probability of reaching a branch as well as its conditional success rate.

既要乘分支内成功率，也要乘到达该分支的概率。

Practice / 配套练习: AIME-2016-II-Q02 (WB p. 756), AIME-2017-II-Q02 (WB p. 757), AIME-2022-II-Q02 (WB p. 759), AIME-2023-II-Q06 (WB p. 772)

Section 5.2 Conditional probability & independence

条件概率与独立性

Method 4 / 方法 4 [PS02-M04]

Normalize independent weights and convolve their distributions

归一独立权重并卷积其分布

Procedure

1. Turn proportional weights into probabilities by dividing by their total.

2. For independent variables, multiply probabilities of elementary coordinate outcomes and sum disjoint ways to reach the target.

3. A probability-generating polynomial encodes a nonnegative integer count; multiply polynomials for independent sums.

4. Matching two independent identical distributions gives $\sum p_k^2$; transformed walk coordinates require proof that their joint one-step signs are uniform.

通用步骤

1. 将成比例权重除以总权重，化为概率。

2. 独立变量的基本坐标结果概率相乘，再对达到目标的互斥方式求和。

3. 概率生成多项式编码非负整数计数；独立和对应多项式相乘。

4. 两个独立同分布计数相等的概率为 $\sum p_k^2$；变换后的随机游走坐标则须证明每步联合符号均匀。

> Example. Independent coins have head probabilities $1/2$ and $1/3$. Find the probability of exactly one head using a generating polynomial.

> 例题：两枚独立硬币正面率为 $1/2, 1/3$，以生成多项式求恰一次正面的概率。

Section 5.2 Conditional probability & independence

条件概率与独立性

Method 4 / 方法 4 [PS02-M04] (continued / 续)

Worked solution

1. The two head-count polynomials are $(1+t)/2$ and $(2+t)/3$.

2. Their product is $(2+3t+t^2)/6$.

3. The coefficient of t is $3/6 = 1/2$, agreeing with $(1/2)(2/3) + (1/2)(1/3)$.

例题解析

1. 两枚硬币的正面计数多项式为 $(1+t)/2, (2+t)/3$。

2. 乘积为 $(2+3t+t^2)/6$。

3. t 的系数为 $3/6 = 1/2$，也等于 $(1/2)(2/3) + (1/2)(1/3)$。

Independence is a property to prove or a stated assumption, not a consequence of naming two variables.

独立性须有已知条件或证明，并非命名两个变量后便自动成立。

Practice / 配套练习: AIME-2006-II-Q05 (WB p. 750), AIME-2010-I-Q04 (WB p. 784), AIME-2014-I-Q02 (WB p. 751), AIME-2014-I-Q11 (WB p. 792), AIME-2016-I-Q02 (WB p. 755), AIME-2023-II-Q06 (WB p. 772)

Section 5.2 Conditional probability & independence

条件概率与独立性

Method 5 / 方法 5 [PS02-M05]

Update the remaining population or use an exchangeable full deal

更新剩余总体，或使用可交换的完整分配

Procedure

1. After each draw, record the remaining multiplicities; without replacement changes later probabilities.

2. Multiply the next-step conditional probabilities along a path and add disjoint draw patterns.

3. If chronological cases are cumbersome, deal all labeled objects in advance and prove permuting batch labels preserves uniformity.

4. Track the number of complete groups left after a mismatch, rather than assume each unobserved pair is equally intact.

通用步骤

1. 每次抽取后记录剩余重数，不放回会改变后续概率。

2. 沿路径乘下一步条件概率，再对互斥抽取模式求和。

3. 若按时间分类繁琐，可先完整分配所有有标签物品，并证明批次标签置换保持均匀性。

4. 不匹配后须追踪剩余完整组数，不能假设所有未见配对仍完整。

> Example. An urn has two red and two blue balls. Draw two without replacement. Show that the events first-red and second-red are not independent.

> 例题：袋中有二红二蓝，不放回抽两球。证明首次红与第二次红两个事件不独立。

Section 5.2 Conditional probability & independence

条件概率与独立性

Method 5 / 方法 5 [PS02-M05] (continued / 续)

Worked solution

1. The first-red probability is $1/2$, and by exchangeability so is the marginal second-red probability.

2. Given the first is red, only one red remains among three balls, so $P(R_2 \mid R_1) = 1/3$.

3. This differs from the marginal $1/2$; equivalently $P(R_1 \cap R_2) = 1/6 \neq 1/4$.

例题解析

1. 首次红概率为 $1/2$，由可交换性，第二次红的边缘概率也为 $1/2$。

2. 已知首次红，剩余三球仅一红，故 $P(R_2 \mid R_1) = 1/3$。

3. 它不同于边缘概率 $1/2$；等价地 $P(R_1 \cap R_2) = 1/6 \neq 1/4$。

Exchangeability of batch labels does not mean independence of the batches.

批次标签可交换，并不意味着各批次独立。

Practice / 配套练习: AIME-2004-II-Q02 (WB p. 749), AIME-2015-I-Q05 (WB p. 753)

Section 5.2 Conditional probability & independence

条件概率与独立性

Method 6 / 方法 6 [PS02-M06]

Condition on the minimal memory state and track parity if needed

按最小记忆状态取条件，必要时追踪奇偶

Procedure

1. Choose the shortest memory that determines the next transition probabilities, such as the previous winner or a matched pattern prefix.

2. For a fixed finite event, multiply transition probabilities along each allowed state path, counting switches rather than assuming independent outcomes.

3. For a stopping parity, let $u_s = E((-1)^R)$ for remaining time R; each next step changes sign, and immediate stopping contributes -1.

4. Solve the first-step equations and convert u to odd probability $(1-u)/2$; establish almost-sure stopping.

通用步骤

1. 选择决定下一步转移概率的最短记忆，例如前次胜者或已匹配模式前缀。

2. 固定有限事件沿各允许状态路径乘转移概率，并统计切换次数，不能假设结果独立。

3. 停时奇偶可设 $u_s = E((-1)^R)$，其中 R 为剩余步数；每前进一步变号，立即停止贡献 -1。

4. 解首步方程，并以 $(1-u)/2$ 转为奇数概率；同时证明几乎必然停止。

> Example. Toss a fair coin until two consecutive heads. Find the probability that the stopping time is odd.

> 例题：投公平硬币，直到连续两次正面，求停止次数为奇数的概率。

Section 5.2 Conditional probability & independence

条件概率与独立性

Method 6 / 方法 6 [PS02-M06] (continued / 续)

Worked solution

1. Use states 0 (no trailing head) and 1 (one trailing head). Let u_0, u_1 be signed remaining-time expectations.

2. First-step conditioning gives $u_0 = -(u_0 + u_1)/2$ and $u_1 = -(u_0 + 1)/2$.

3. Thus $u_1 = -3u_0$ and $u_0 = 1/5$, so the odd probability is $(1 - 1/5)/2 = 2/5$.

4. Each disjoint two-toss block has probability $1/4$ of being HH, so the chance of avoiding a stop forever is zero.

例题解析

1. 取状态 0（无末尾正面）与 1（末尾一个正面），设 u_0, u_1 为带符号剩余时间期望。

2. 首步取条件得 $u_0 = -(u_0 + u_1)/2, u_1 = -(u_0 + 1)/2$。

3. 故 $u_1 = -3u_0, u_0 = 1/5$，奇数概率为 $(1 - 1/5)/2 = 2/5$。

4. 每个互不相交的两投块有 $1/4$ 概率为 HH，故永远不停止的概率为零。

A failed prefix may retain a shorter matching suffix; recalculate the state rather than always resetting to zero.

前缀失败后可能保留较短匹配后缀，应重新确定状态，不能总是归零。

Practice / 配套练习: AIME-2018-II-Q13 (WB p. 780), AIME-2021-I-Q01 (WB p. 758)